

2023 FIFA Women's World Cup Shots

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Player	Shot Outcome	Play Pattern	Under Pressure	Distance to Goal (Yards)
Elizabeth Doon Hassett	Blocked	Regular Play	TRUE	14.55
Ada Hegerberg	Blocked	Regular Play	FALSE	30.51
Ada Hegerberg	Blocked	From Counter	TRUE	12.06
Hannah Wilkinson	Blocked	From Counter	TRUE	13.22
Hannah Wilkinson	Goal	From Goal Kick	FALSE	7.34
Source: StatsBomb				

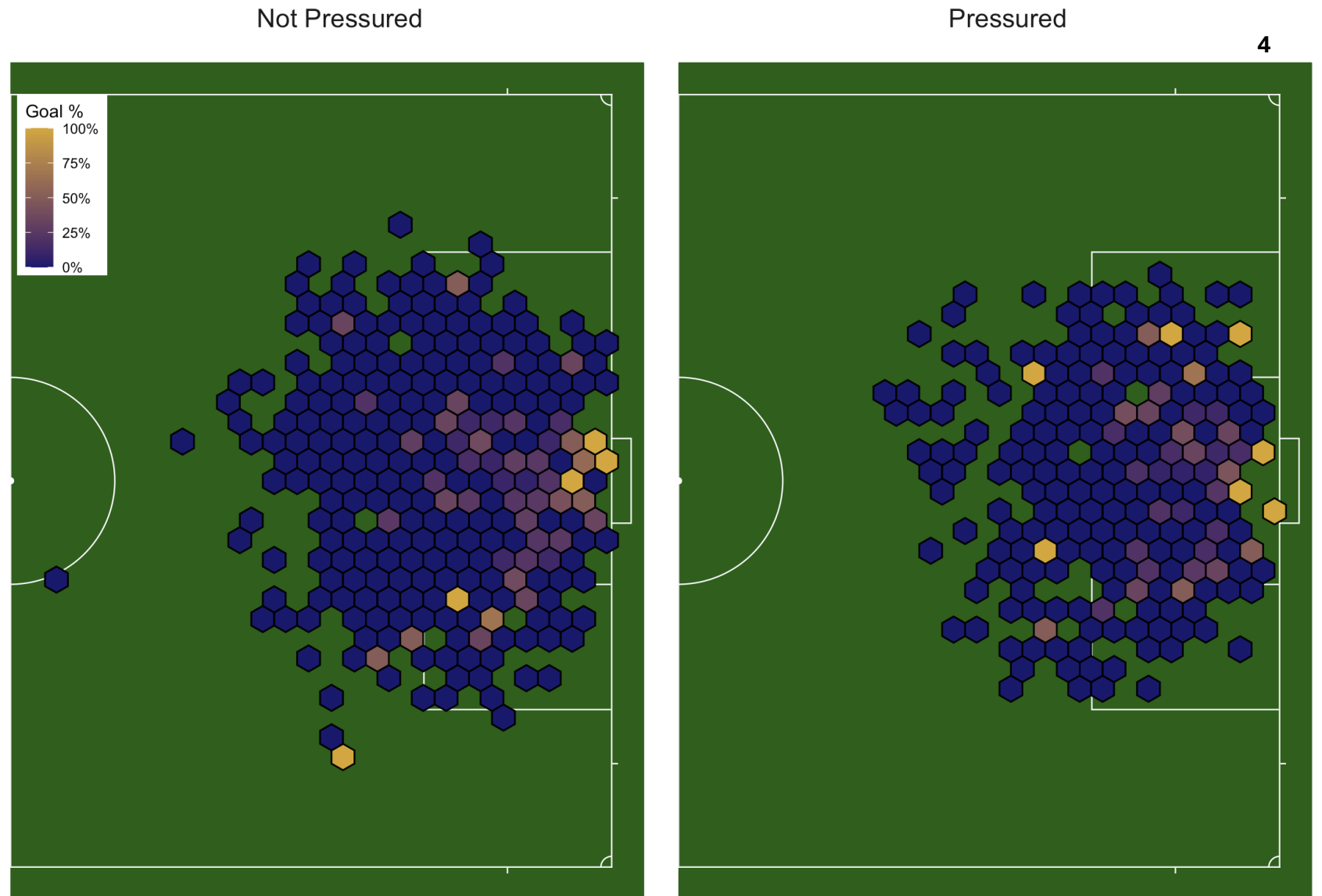
Data: Shot-level data from the 2023 Women's World Cup

Question 1: Does defensive pressure influence shot success?



Hypothesis: Shots pressured by defenders are less likely to result in goals.

Defensive
pressure **did**
not impact
goal rate



Data courtesy of StatsBomb

	Goals	Shots	Goal %
Not Pressured	77	886	8.69%
Pressured	56	658	8.51%

Question 1 Conclusion

- 42.6% of shot attempts in the data were under pressure.
- Observed a 0.018% rise in goal rate when not pressured by defenders.



2023

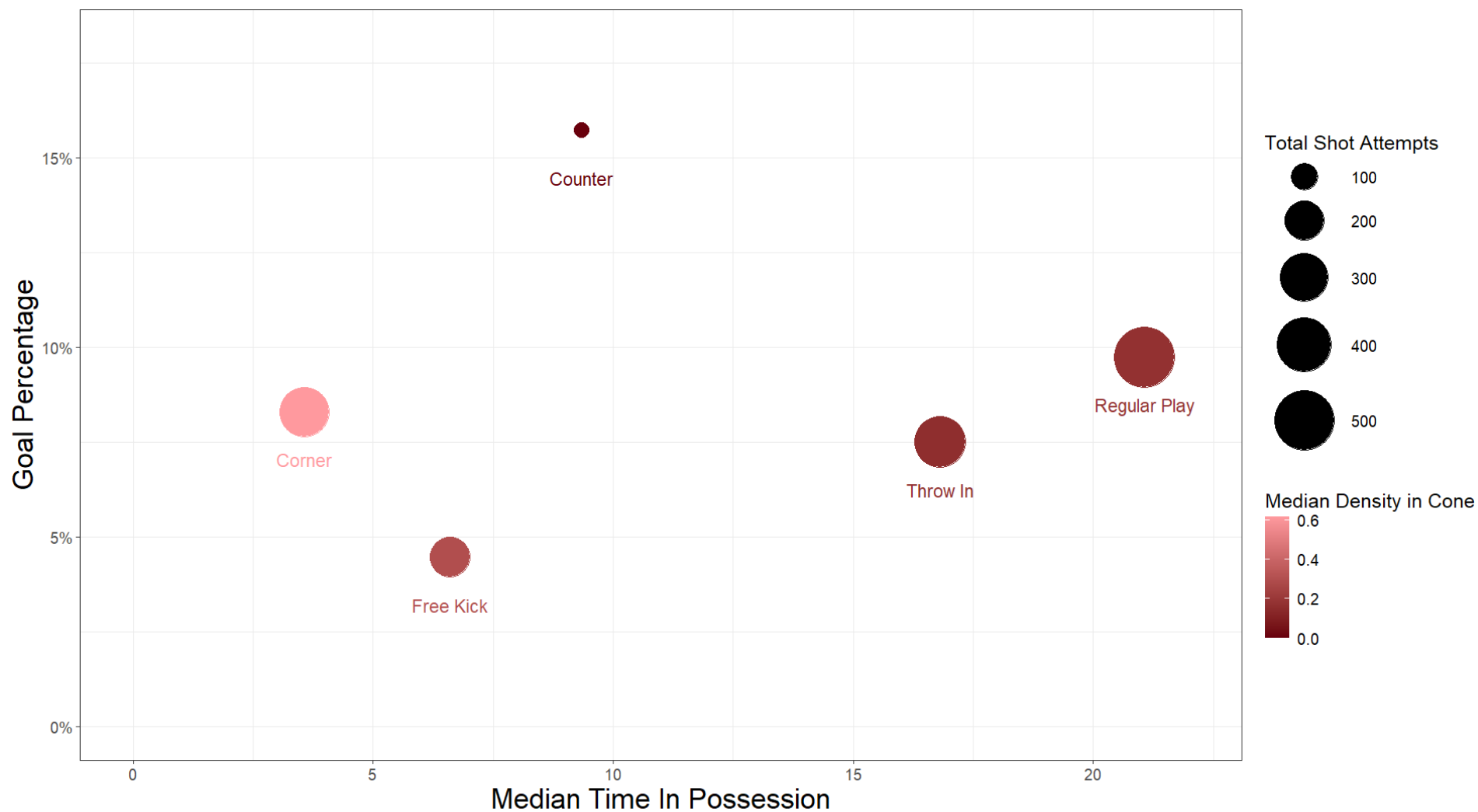
Question 2: What specific play patterns create dangerous shots?



Hypothesis: Counter-shot attempts and shots quickly after established team possession have fewer defenders in the cone and higher goal rates than regular, established-possession attacks.

Counter shots produced the highest goal rate

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Data courtesy of StatsBomb

Question 2 Conclusion

- Counter shot attempts generated the highest goal percentage ($>15\%$) and the lowest median defender density in the cone out of the most common play patterns.
- Regular plays and throw-ins had the highest median possession time leading up to the shot attempt.



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Identifying What Made the Top 15 Scorers Successful

We performed **gaussian mixture clustering** with:

1. Shot distance to the goal
2. Average velocity

Performed with the **top 15** scorer's shot attempts *only*

Component	Size	Proportion	Avg Distance To Goal	Avg Velocity
1	57	0.28	7.18	9.83
2	126	0.72	15.49	16.23

Clustering Conclusion

- The best goal scorers from the tournament do not need to be close to the goal to find success.
- Top goal scorers rely on ball speed to beat a goalkeeper.



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Discussion

Limitations

- Excluded shots with velocities > 30 yd/s (not realistic, possible data entry error)
- Tournament style data skews towards the best teams

Future Work

- Shooting Talent Model: Do two different players shooting under the same circumstances (distance, pressure, etc.) have the same probability of resulting in a goal?

